Small circuits can have exponentially large prior complexity below the mean-field scaling

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Abstract

We study a fully connected Bayesian tanh network on the Boolean cube, with Gaussian layer variances on the scaling line indexed by γ, and a prior conditioned to have a fixed output margin at every input. For every fixed $\gamma < 1$, we show that a small circuit need not have polynomial prior complexity. Full parity on d bits has a depth-L implementation with width $d+1$ and weights $O(d \log d)$, for every $L \geq 2$. Nevertheless, for $\gamma \leq 0$ its limiting normalized prior probability is at most 2^{-2^d}. For $0 < \gamma < 1$, its negative log prior probability, divided by N^γ, is at least $a^2(2^d - O_L(d))/2$ in the width limit, where a is the margin. The proof uses the exact spectrum of the saturated neural-network kernel and a general Gaussian inequality: a Boolean minimum-eigenvalue eigenvector has probability at most 2^{-M} among sign patterns, even after conditioning on any common global margin. These are prior-mass separations. They do not by themselves establish exponential sample complexity for the resulting Bayesian classifier.

1 Model and the prior-complexity question

Let $\mathcal{X} = \{-1, 1\}^d$, $M = 2^d$, and $L \geq 2$. The input is augmented by a constant coordinate, $h^0(x) = (1, x)$. A network with $L-1$ hidden layers of common width N has

$$h^s(x) = \tanh(W^s h^{s-1}(x)), \qquad z_N(x) = W^L h^{L-1}(x).$$

The readout is linear, and there are no further bias coordinates. All weights are independent centered Gaussians with variances

$$\mathrm{Var}(W^1_{ij}) = N^{1-\gamma}/d, \qquad \mathrm{Var}(W^s_{ij}) = N^{-\gamma} \quad (1 < s < L), \qquad \mathrm{Var}(W^L_i) = N^{-\gamma-1}. \tag{1}$$

Fix $a > 0$. Write $Z_N = (z_N(x))_{x \in \mathcal{X}}$, and define

$$A_a = \{|Z_N(x)| > a \text{ for every } x\}, \qquad A_f = \{f(x) Z_N(x) > a \text{ for every } x\}.$$

The prior on Boolean functions is

$$\pi_{N,a}(f) = \mathbb{P}(A_f)/\mathbb{P}(A_a). \tag{2}$$

Thus the all-input margin restriction is applied before observing any training data. It is not a margin imposed only on the sample.

Let $C_L(f)$ be the smallest integer C such that a depth-L network of width at most C, with all weights of magnitude at most C, satisfies $f(x)z(x) > 1/2$ everywhere. This combines width and weight size in one circuit-complexity parameter.

The natural prior complexity changes with the overall output scale. For the present paper define

$$\mathcal{C}_{\gamma,a}(f) = \begin{cases} \lim\limits_{N\to\infty} -\log \pi_{N,a}(f), & \gamma \le 0, \\ \lim\limits_{N\to\infty} -N^{-\gamma} \log \pi_{N,a}(f), & 0 < \gamma < 1. \end{cases} \tag{3}$$

We prove below that these limits exist. The normalization for positive γ removes the Gaussian small-noise speed. Without it, negative log probabilities would diverge even for many simple functions.

All width limits take $N \to \infty$ at fixed d, L, γ, a. Statements about growth in d concern the resulting limiting quantities. No uniform exchange of the width and dimension limits is made.

Theorem 1.1 (A circuit-prior separation throughout $\gamma < 1$)**.** *Let $f_d(x) = \prod_{j=1}^{d} x_j$ be full parity, and put $c_L = (2/\pi)^{L-1}$. For every $L \ge 2$,*

$$C_L(f_d) \le \lceil (d+1)\log(16(d+1)) \rceil . \tag{4}$$

For every $\gamma \le 0$,

$$\mathcal{C}_{\gamma,a}(f_d) \ge (\log 2) 2^d. \tag{5}$$

For every $0 < \gamma < 1$,

$$\mathcal{C}_{\gamma,a}(f_d) \ge \frac{a^2}{2}\left(2^d - \frac{d+1}{c_L}\right). \tag{6}$$

Consequently, at each fixed depth and fixed subcritical exponent, there is no polynomial in d and $C_L(f)$ that upper-bounds $\mathcal{C}_{\gamma,a}(f)$ for every Boolean target f.

This theorem answers a prior-mass question: throughout this scaling regime, polynomial-size circuits need not receive polynomial description-length prior mass, on the natural scale of the prior. The distinction between this statement and a posterior learning lower bound is discussed in Section 6.

2 The saturated kernel and its spectrum

Put

$$H(t) = \frac{2}{\pi}\arcsin t, \qquad F = H^{\circ(L-1)}, \qquad K(x,y) = F\left(\frac{1 + x\cdot y}{d+1}\right). \tag{7}$$

For $U \subseteq \{1,\dots,d\}$, write $\chi_U(x) = \prod_{j\in U} x_j$. Translation symmetry on the Boolean cube makes these parity functions eigenvectors of K. Let λ_r denote the eigenvalue of the normalized operator $M^{-1}K$ on the degree-r parities.

Lemma 2.1 (Paired ordering of the eigenvalues)**.** *The matrix K is positive definite, and*

$$\lambda_0 = \lambda_1 > \lambda_2 = \lambda_3 > \lambda_4 = \lambda_5 > \cdots > 0, \tag{8}$$

with a final unpaired λ_d if d is even. Moreover,

$$\lambda_d \le 2^{-d}, \qquad \lambda_0 \ge \frac{c_L}{d+1}. \tag{9}$$

Proof. Write $D = d + 1$. The arcsine power series has positive coefficients at every odd degree, zero even coefficients, and sum one at $t = 1$. These properties persist under composition. Thus $F(t) = \sum_{m \text{ odd}} c_m t^m$, where every $c_m > 0$, $\sum_m c_m = 1$, and $c_1 = c_L$.

First use the full D-dimensional cube and the kernel $F(u \cdot v/D)$. The degree-r Fourier coefficient of $(u \cdot v/D)^m$ equals $p_m(r)$, the probability that the simple coordinate-flip walk reaches a specified vertex at distance r after m steps. Indeed, a term in the power expansion survives Fourier averaging exactly when the coordinates chosen an odd number of times form that specified support.

For $0 \le r \le D - 2$, the walk recursion gives

$$\begin{aligned} D[p_{m+1}(r) - p_{m+1}(r+2)] = r[p_m(r-1) - p_m(r+1)] \\ + (D - r - 2)[p_m(r+1) - p_m(r+3)]. \end{aligned}$$

Terms with zero coefficients at the boundary are omitted. Starting from a point mass at time zero, induction proves $p_m(r) \ge p_m(r+2)$. The full-cube eigenvalues $\mu_r = \sum_{m \text{ odd}} c_m p_m(r)$ vanish for even r and strictly decrease through the odd values of r: for odd r, the term $m = r$ is positive at distance r and zero at distance $r + 2$. Each odd μ_r is positive.

Restricting the distinguished coordinate to $+1$ gives $\lambda_r = \mu_r + \mu_{r+1}$. This proves (8) and positive definiteness. Since $K(x, x) = 1$,

$$\sum_{r=0}^{d} \binom{d}{r} \lambda_r = 1.$$

Its smallest eigenvalue therefore satisfies $\lambda_d \le 1/M$. Finally, the linear term of F contributes $c_L/(d+1)$ to λ_0, and all other contributions are nonnegative. This proves the remaining bound. □

3 A Gaussian sign inequality that survives global conditioning

Lemma 3.1 (Minimum-eigenvalue signs have at most uniform mass)**.** *Let Σ be a positive-definite $M \times M$ covariance matrix, and let $f \in \{-1, 1\}^M$ be an eigenvector for its smallest matrix eigenvalue $v > 0$. If $G \sim \mathcal{N}(0, \Sigma)$, then for every $a \ge 0$,*

$$\mathbb{P}\{f_i G_i > a \text{ for every } i\} \le \Phi(-a/\sqrt{v})^M, \tag{10}$$

and

$$\mathbb{P}\{\operatorname{sign} G = f \mid |G_i| > a \text{ for every } i\} \le 2^{-M}. \tag{11}$$

Here Φ is the standard normal distribution function.

Proof. Conjugate the covariance by $\operatorname{diag}(f)$, so the target sign is the all-ones vector $\mathbf{1}$ and this vector has eigenvalue v. Write the conjugated Gaussian as

$$G' = \sqrt{v}\,\xi + Y,$$

where ξ has independent standard normal coordinates and Y is independent with covariance $\Sigma' - vI$. This covariance is positive semidefinite and kills $\mathbf{1}$. Consequently $\sum_i Y_i = 0$ almost surely. Set $\alpha = \Phi(-a/\sqrt{v})$.

The function $\log \Phi$ is concave. Conditional on Y, Jensen's inequality therefore gives

$$\mathbb{P}\{G_i' > a \ \forall i \mid Y\} = \prod_{i=1}^{M} \Phi\left(\frac{Y_i - a}{\sqrt{v}}\right) \le \alpha^M.$$

Integrating proves (10). To verify the concavity directly, put $r(t) = \phi(t)/\Phi(t)$. Then $(\log \Phi)''(t) = -r(t)(t + r(t)) < 0$. For $t < 0$ the last factor is positive because $\phi(t) = \int_{-\infty}^{t} (-s)\phi(s)\, ds > -t\Phi(t)$; for $t \geq 0$ it is immediate.

A one-dimensional Gaussian with variance v is least likely to lie outside $[-a, a]$ when its mean is zero. Indeed the outside probability is even in the mean, and its derivative is nonnegative for nonnegative means. Thus, conditional on Y,

$$\begin{aligned}\mathbb{P}\{|G_i'| > a \ \forall i \mid Y\} &= \prod_{i=1}^{M} \left[\Phi\left(\frac{Y_i - a}{\sqrt{v}}\right) + \Phi\left(\frac{-Y_i - a}{\sqrt{v}}\right)\right] \\ &\geq (2\alpha)^M.\end{aligned}$$

Integrating and dividing the numerator and denominator bounds gives (11). Conjugation preserves the global absolute-margin event, completing the proof. □

The lemma does not require a quantitative lower bound on v or an estimate of the global margin probability. In particular it does not lose a large factor when the conditioning event is rare.

4 Width limits and the prior-rate calculation

We include the limiting argument to specify exactly how the Gaussian calculation applies to the finite networks.

Proposition 4.1 (Kernel freezing and output limits)**.** *Fix $\gamma < 1$. Let Q_N be the empirical covariance of the final hidden features on the full cube. For every $\epsilon > 0$, there are $c, C > 0$ and N_0 such that*

$$\mathbb{P}\{\|Q_N - K\|_{\mathrm{op}} > \epsilon\} \leq Ce^{-cN} \qquad (N \geq N_0). \tag{12}$$

Conditional on the hidden layers,

$$Z_N \sim \mathcal{N}(0, N^{-\gamma} Q_N). \tag{13}$$

For $\gamma = 0$, the globally margin-conditioned sign prior converges to that of $G \sim \mathcal{N}(0, K)$ with margin a. For $\gamma < 0$, it converges to the unconditioned sign law of G. For $0 < \gamma < 1$, Z_N has a large-deviation principle at speed N^γ with rate $I(z) = z^T K^{-1} z/2$.

Proof. Put $b_N = N^{(1-\gamma)/2} \to \infty$. Conditional on a hidden layer, the next preactivation rows are independent Gaussian vectors with covariance b_N^2 times that layer's empirical covariance. The first layer has the deterministic covariance $b_N^2(1 + x \cdot y)/d$, whose correlation matrix is $(1 + x \cdot y)/(d + 1)$.

If $U \sim \mathcal{N}(0, q)$ with q bounded below, direct integration gives

$$\mathbb{E}|\tanh(bU) - \operatorname{sign} U| \leq \frac{2 \log 2}{b\sqrt{2\pi q}}.$$

The correlation of the signs of a centered Gaussian pair with correlation r is $H(r)$. Thus conditional expected products of hidden activations approach the normalized arcsine transform, uniformly in a sufficiently small neighborhood of each limiting covariance. All distinct-input correlations stay strictly between -1 and 1, and diagonal variances stay bounded below. The off-diagonal normalized arcsine map is consequently locally Lipschitz there; the limiting diagonal is identically one.

For each pair of inputs the N conditional row products lie in $[-1, 1]$ and are independent. Hoeffding's inequality and a union bound over the fixed M^2 pairs give exponential concentration

at speed N. Induction over the fixed number of layers, absorbing the deterministic $O(b_N^{-1})$ error, proves (12). The independent Gaussian readout gives (13).

It follows that $N^{\gamma/2}Z_N$ converges weakly to the nondegenerate Gaussian G. At $\gamma = 0$, all margin orthants and their unions have boundary of Gaussian measure zero and positive probability. Their conditional probabilities therefore converge. At $\gamma < 0$, the normalized margin is $aN^{\gamma/2} \to 0$. The probability that any coordinate lies in this vanishing interval tends to zero, giving the asserted sign limit.

For $0 < \gamma < 1$, couple

$$Z_N = N^{-\gamma/2} Q_N^{1/2} \xi, \qquad Y_N = N^{-\gamma/2} K^{1/2} \xi,$$

where $\xi \sim \mathcal{N}(0, I_M)$ is independent of Q_N. The square-root map is locally Lipschitz near the positive-definite K. For fixed $\delta, \epsilon > 0$, this gives

$$\mathbb{P}\{\|Z_N - Y_N\| > \delta\} \le C_\epsilon e^{-c_\epsilon N} + \mathbb{P}\left\{\|\xi\| > \frac{\delta N^{\gamma/2}}{C\epsilon}\right\}.$$

Divide logarithms by N^γ, let $N \to \infty$, and then let $\epsilon \downarrow 0$. The upper limit is $-\infty$, since $\gamma < 1$. This is exponential equivalence at speed N^γ. The Gaussian family Y_N has the claimed quadratic rate, and exponential equivalence transfers its large-deviation principle. □

Proposition 4.2 (Exact normalized parity rate)**.** *For $0 < \gamma < 1$,*

$$\mathcal{C}_{\gamma,a}(f_d) = \frac{a^2}{2}\left(\frac{1}{\lambda_d} - \frac{1}{\lambda_0}\right). \tag{14}$$

Proof. For a Boolean sign function g, define

$$E_K(g) = \min\left\{\frac{1}{2} z^T K^{-1} z : g(x)z(x) \ge a \text{ for all } x\right\}.$$

The feasible set is closed and the objective strictly convex and coercive, so it has a unique minimizer. Scaling that minimizer by $1 + \eta$ puts it in the strict margin orthant and approaches its energy as $\eta \downarrow 0$. The open lower and closed upper bounds in Proposition 4.1 therefore agree:

$$\lim_{N\to\infty} -N^{-\gamma} \log \mathbb{P}(A_g) = E_K(g).$$

The global event is the disjoint union of finitely many such orthants. Its rate is $\min_g E_K(g)$, so the normalized prior rate is $E_K(g) - \min_h E_K(h)$.

If a Boolean vector g has normalized eigenvalue λ, then $z = ag$ minimizes its margin energy. Its gradient is $K^{-1}(ag) = ag/(M\lambda)$, whose inner product with every feasible displacement is nonnegative. Hence $E_K(g) = a^2/(2\lambda)$.

For any full-margin vector z,

$$\frac{1}{2} z^T K^{-1} z \ge \frac{\|z\|_2^2}{2M\lambda_0} \ge \frac{a^2}{2\lambda_0}.$$

The constant sign attains equality. Full parity has eigenvalue λ_d, proving (14). □

For $\gamma \le 0$, the limiting sign probabilities are positive, so negative logarithms converge as in (3). Applying Lemma 3.1 to full parity and the appropriate margin proves (5). Combining Proposition 4.2 with Lemma 2.1 proves (6).

5 An explicit small circuit for parity

Proposition 5.1 (Alternating Hamming-weight thresholds)**.** *For every $d \geq 1$ and $L \geq 2$, there is a depth-L* tanh *network of width $d+1$, using only the appended input bias, whose output approximates f_d uniformly within $1/8$ and whose weights have magnitude at most $(d+1)\log(16(d+1))$.*

Proof. Set $A = \log(16(d+1))$, $s(x) = \sum_i x_i$, and for $1 \leq j \leq d$ put

$$t_j = -d + 2j - 1, \qquad q_j(x) = \operatorname{sign}(s(x) - t_j), \qquad c_j = (-1)^{d-j}.$$

Let $\alpha_d = 1$ for even d and $\alpha_d = 0$ for odd d. The exact identity

$$f_d(x) = \alpha_d + \sum_{j=1}^{d} c_j q_j(x) \tag{15}$$

follows by writing r for the number of $+1$ bits. Then $q_j = +1$ exactly when $j \leq r$. At $r = 0$ both sides equal $(-1)^d$, and each increment of r changes both sides by $2(-1)^{d-r}$.

The first hidden layer contains $h_j^1(x) = \tanh(A(s(x) - t_j))$ for $1 \leq j \leq d$, together with the constant neuron $h_0^1(x) = \tanh A$. Each $s(x) - t_j$ is a nonzero odd integer. All these activations consequently have their desired sign and differ from it by at most $\delta = 1/[8(d+1)]$.

For every additional hidden layer use diagonal weights A and zero off-diagonal weights. If an activation has the correct sign and magnitude at least $1/2$, its successor differs from that sign by at most $2e^{-A} = \delta$. Thus the same approximation holds at every hidden layer, without accumulated error.

Use the linear readout

$$z(x) = \alpha_d h_0^{L-1}(x) + \sum_{j=1}^{d} c_j h_j^{L-1}(x).$$

By (15), its uniform error is at most $(d+1)\delta = 1/8$. In particular its correct margin is at least $7/8$. The largest weight is at most $A\max(1, d-1)$, including the first-layer threshold biases. This is bounded by $(d+1)A$. □

This proves (4) and completes the proof of Theorem 1.1. If a circuit implementing an arbitrary fixed margin a is desired, rescale the readout by a constant depending only on a; the polynomial size conclusion is unchanged.

6 What remains a learning question

The theorem separates circuit complexity from prior complexity throughout $\gamma < 1$, including $k = d$ for a k-bit XOR. It does not establish that the actual posterior requires exponentially many examples to achieve small test error or correct-label probability strictly greater than $2/3$. A low prior probability can be overcome by a small sample if that sample eliminates the competing functions.

For example, consider the Gaussian process $G(x) = A + Bf(x)$, where A and B are independent nondegenerate centered Gaussians and f is a nonconstant parity. Even after any global margin conditioning, its possible sign functions are only $1, -1, f, -f$. Once examples with both target labels have been seen, f is the unique compatible sign function and every prediction has correct-label probability one. The variance of B can be arbitrarily small, making the initial probability of f arbitrarily unfavorable without changing that conclusion. This example does not contradict Lemma 3.1: its covariance is singular, and f is not a minimum-eigenvalue eigenvector.

For the networks studied here, the same kernel limit identifies the posterior completely at $\gamma \leq 0$ and its leading energy at $0 < \gamma < 1$. At $\gamma \leq 0$ it is a Gaussian sign classifier with the margin specified in Proposition 4.1. At $0 < \gamma < 1$, conditioned on observed training signs, its mass concentrates on compatible sign functions minimizing E_K. The optimization retains $|z(x)| \geq a$ at every unseen input. Its leading optimization uses a fixed kernel and a nonconvex union of full margin orthants as its feasible set. This statement does not determine posterior weights between tied minimizers. In fact, the companion note *Gaussian conditioning below the mean-field scaling: a sharp fluctuation threshold* gives an actual-network example, with ties not related by a cube or weight symmetry, where non-Gaussian corrections change those weights and a 2/3-confidence prediction for $1/2 \leq \gamma < 1$. Ordinary Gaussian-process regression or a hard-margin support vector machine using only the observed inputs is a different rule.

An exponential sample lower bound for this precise classifier requires control of compatible competitors and posterior marginal sign probabilities, beyond the prior separation proved here. The confidence criterion $> 2/3$ makes that distinction especially important: a tiny posterior mass on the exact target function can coexist with highly confident correct marginal predictions at every input.